NEURALSPARK

System Architecture & Multimodal Infrastructure Specification

4-Tier Microservices, Multimodal Handwritten PDF Grading Pipeline, and Edge Infrastructure

Project Team:

Malay Singh (2303031249003) **Prashant Nagre** (2303031240687)

Shreyansh Negi (2303031241262) **Nitesh Sukhwal** (2303031240866)

Project Guide: **Prof. RASNA PATEL** (AI/DS Department, PIET)

Department of Artificial Intelligence and Data Sciences

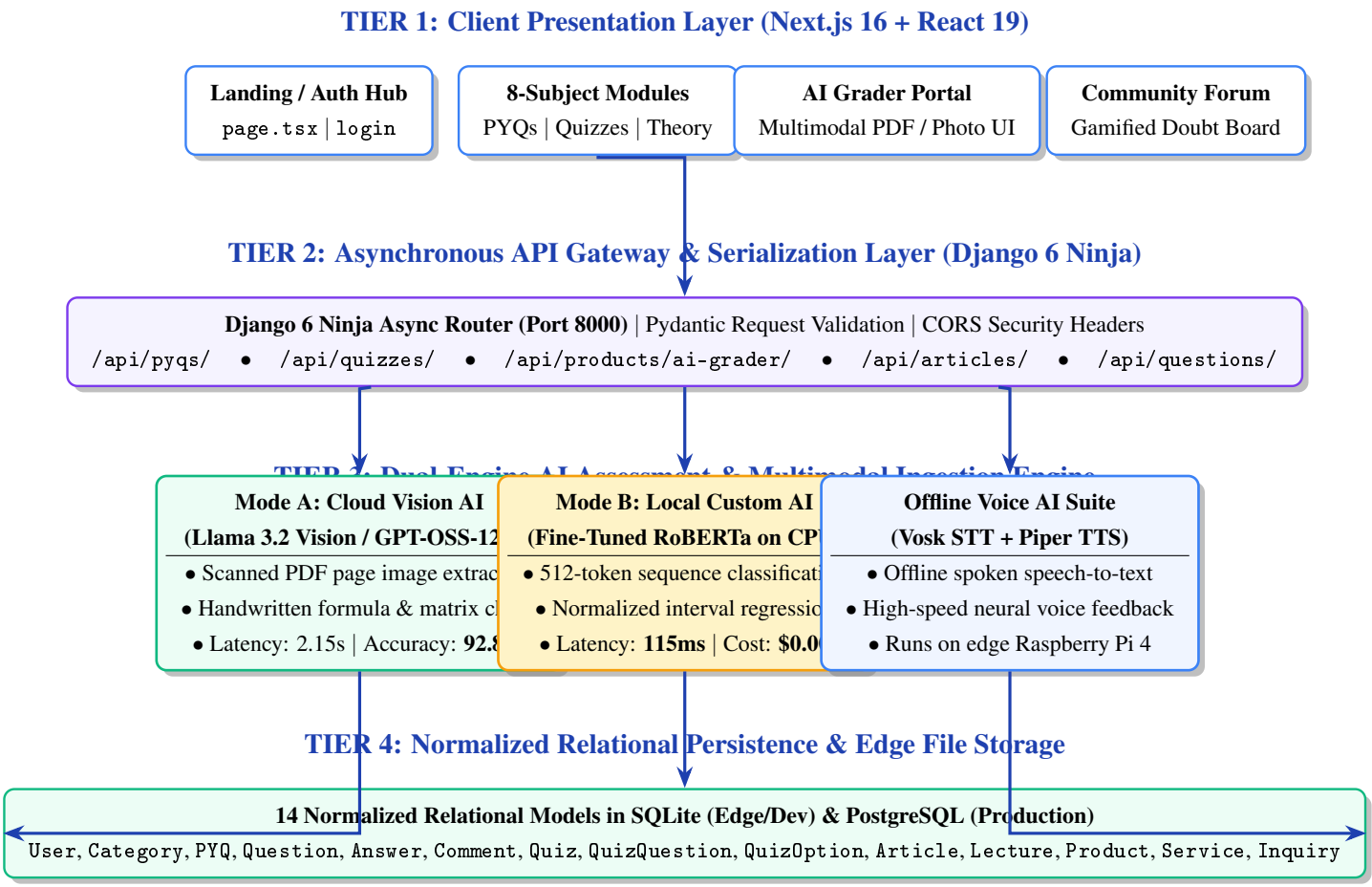
Parul Institute of Engineering and Technology, Parul University, Vadodara

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1 1. 4-Tier Microservices Architecture Overview

NeuralSpark is designed as a decoupled, multi-tiered microservices architecture optimized for sub-35ms REST response times, high-accuracy multimodal visual grading, and zero-connectivity offline edge resilience.



2 2. Multimodal Handwritten PDF & Visual Ingestion Pipeline

This section details how the multimodal engine ingests, extracts, sanitizes, and evaluates scanned handwritten documents and camera photos.

1. Student Assignment Upload Interface:

2. Multipart Extraction Pipeline (`_process_uploaded_file`):

- **Digital Text Stream:** Extracted via `PyPDF2.extract_text()`
- **Scanned Page Bitmap Stream:** Iterates `page.images`, extracts raw bytes, encodes to Base64 (`data:image/jpeg;base64`)
- **Hybrid Assembly:** Combines digital text tokens and visual image payloads into a unified OpenAI/Groq-

3. Grounded System Prompt Injection:
Enforces strict Teaching Assistant persona, step-by-step mathematical reasoning, line-by-line equation checks, and structured `Score: X/Y` output format

4. High-Capacity Vision LLM Execution (Llama 3.2 Vision / GPT-OSS-120B via Groq API):
Visual character recognition → Matrix determinant step check → Factored polynomial check → Mark deduction justification

5. Client Serialization & Persistence:
JSON response returned to Next.js in < 2.5 seconds → Stored in Database → Rendered with animated score reveal and remedial guidance

3 3. System Architectural Component Specifications

Layer / Component	Technology Stack	Key Protocols / Format	Architectural Role
Frontend Client	Next.js 16, React 19, TypeScript	HTTP/2, Server Components, JSON	Ultra-responsive UI, KaTeX math rendering, theme persistence
API Gateway	Django 6 Ninja Async	REST, Pydantic, CORS	Sub-35ms routing, schema validation, async coroutine handlers
Cloud Vision AI	Llama 3.2 Vision / GPT-OSS	Base64 Multimodal JSON	Handwritten PDF and diagram grading with 92.8% accuracy
Local Offline AI	RoBERTa / BERT (PyTorch)	CPU Tensor Classification	Zero-cost, sub-120ms private sequence grading without internet
Speech Pipeline	Vosk STT + Piper TTS	16kHz PCM WAV Audio	Offline edge voice tutoring in rural connectivity-limited classrooms
Database Layer	SQLite (Edge) / PostgreSQL	Django ORM, SQL Indexes	14 normalized relational entities with foreign-key referential integrity

Complete source code, models, and documentation available at:
`https://github.com/malay336/NeuralSpark.git`